

# **TOPIC 08**

## **INPUT AND OUTPUT FILE**

Fundamentals of Computing  
(FSPK0022)

# Introduction to Files

# File Input and Output

- ✿ Can use files instead of keyboard and monitor screen for program input and output.
- ✿ File: a set of data stored on a computer, often on a disk drive.
  - ◆ Allows data to be retained between program runs.
- ✿ Programs can read from and/ or write to files.
- ✿ Used in many applications: word processing, databases, spreadsheets, compilers.
- ✿ Steps: (1) Open the file (2) Use the file (read from, write to, or both) (3) Close the file.

# File Operations

✿ Requires **fstream** header file:

- ◆ use ifstream data type for input files.
- ◆ use ofstream data type for output files.
- ◆ use fstream data type for both input, output files.

✿ **ifstream:**

- ◆ Open for input only and file cannot be written to.
- ◆ Open fails if file does not exist.

✿ **ofstream:**

- ◆ Open for output only and file cannot be read from.
- ◆ File created if no file exists.
- ◆ File contents erased if file exists.

# File Operations (cont.)

✿ **fstream** object can be used for either input or output.

✿ **fstream**: must specify mode on the open statement. Sample modes:

- ◆ **ios::in** for input mode.
- ◆ **ios::out** for output mode.
- ◆ **ios::binary** for binary mode.
- ◆ **ios::app** for append mode. All output operations are performed at the end of the file, appending the content to the current content of the file.

# Opening Files

✿ Create a link between file name (outside the program) and file stream object (inside the program).

✿ Filename may include drive and/or path info.

✿ **ifstream** and **ofstream** - use the **open()** member function:

```
infile.open("inventory.dat");  
outfile.open("report.txt");
```

✿ **fstream** - use the **open()** member function and **mode(s)**:

```
infile.open("inventory.dat", ios::in);  
outfile.open("report.txt", ios::out);
```

# Opening Files (cont.)

✿ **fstream** - can be combined on open call:

```
dFile.open("class.txt", ios::in | ios::out);
```

✿ Can open file at declaration:

```
ifstream gradeList("grades.txt");  
fstream infile("inventory.dat", ios::in);  
fstream file("class.txt", ios::in | ios::out);
```

✿ Output file will be created if necessary; existing file will be erased first.

✿ Input file must exist for open to work.

# Opening Files (cont.)

🌸 File stream object set to **0(false)**, if **open failed**. Example:

```
if (!input)
{ cout << "ERROR: Cannot open file\n";
  exit(1); }
```

🌸 Can use **fail()** member function to detect file open error:

```
if (input.fail())
{ cout << "ERROR: Cannot open file\n";
  exit(1); }
```

🌸 Can use **is\_open()** member function to check if a file is open:

```
if (!input.is_open())
{ cout << "ERROR: Cannot open file\n";
  exit(1); }
```



# Using Files

✿ Can use **output file object** and **<<** to send data to a file:

```
outfile << "Inventory report";
```

✿ Can use **input file object** and **>>** to copy data from file to variables:

```
infile >> partNum;
```

```
infile >> qtyInStock >> qtyOnOrder;
```

✿ Can use **eof()** member function **to test for end of input file.**

# Closing Files

🌸 Use the **close()** member function:

```
infile.close();  
outfile.close();
```

🌸 Don't wait for operating system to close files at program end:

- ◆ may be limit on number of open files.
- ◆ may be buffered output data waiting to send to file.

# Example 1: File Operations

```
#include <iostream> //copy 10 numbers between files
#include <fstream>
using namespace std;

int main()
{
    fstream infile("input.txt", ios::in); // open the files
    fstream outfile("output.txt", ios::out);
    int num;

    for (int i = 1; i <= 10; i++) {
        infile >> num;    // use the files
        outfile << num; }

    infile.close();    // close the files
    outfile.close();

}
```

# Example 2: File Operations

## Program 3-28

```
1 // This program writes data to a file.
2 #include <iostream>
3 #include <fstream>
4 using namespace std;
5
6 int main()
7 {
8     ofstream outputFile;
9     outputFile.open("demofile.txt");
10
11     cout << "Now writing information to the file.\n";
12
13     // Write 4 great names to the file
14     outputFile << "Bach\n";
15     outputFile << "Beethoven\n";
16     outputFile << "Mozart\n";
17     outputFile << "Schubert\n";
18 }
```

## Program 3-28 (continued)

```
19 // Close the file
20 outputFile.close();
21 cout << "Done.\n";
22 return 0;
23 }
```

### Program Screen Output

Now writing data to the file.  
Done.

### Output to File demofile.txt

Bach  
Beethoven  
Mozart  
Schubert

# Example 3: File Operations

## Program 3-29

```
1 // This program reads information from a file.
2 #include <iostream>
3 #include <fstream>
4 using namespace std;
5
6 int main()
7 {
8     ifstream inFile;
9     const int SIZE = 81;
10    char name[SIZE];
11
12    inFile.open("demofile.txt");
13    cout << "Reading information from the file.\n\n";
14
15    inFile >> name;           // Read name 1 from the file
16    cout << name << endl;     // Display name 1
17
18    inFile >> name;           // Read name 2 from the file
19    cout << name << endl;     // Display name 2
20
21    inFile >> name;           // Read name 3 from the file
22    cout << name << endl;     // Display name 3
23
```

## Program 3-29 (continued)

```
24    inFile >> name;           // Read name 4 from the file
25    cout << name << endl;     // Display name 4
26
27    inFile.close();           // Close the file
28    cout << "\nDone.\n";
29    return 0;
30 }
```

## Program Screen Output

Reading data from the file.

Bach  
Beethoven  
Mozart  
Schubert

Done.

# Example 4: File Operations

```
#include <fstream>
using namespace std;
int main()
{
    ifstream input("inputfile.txt");
    char str[80];

    if (!input)
    {
        cout << "While opening a file an error is encountered" << endl;
        return 0;
    }
    else
        cout << "File is successfully opened" << endl;
    while(!input.eof())
    {
        input.getline(str, 80);
        cout << str << endl;
    }
    input.close();
    return 0;
}
```

# Example 5: File Operations

```
#include <iostream>
#include <fstream>
using namespace std;
int main ()
{
    int num;
    ifstream inp("input.txt"); // open the input file
    ofstream out("output.txt"); // open the output file
    if (!inp.is_open()) // check for successful opening
    {
        cout << "Input file could not be opened! Terminating!\n";
        return 0;
    }
    while (inp >> num)
        out << num * 2 << endl;

    inp.close();
    out.close();
    cout << "Done!" << endl;
    return 0;
}
```